



GATE ME 90 Days Challenge

Day 8/90 · Practice Set with Detailed Solutions

SUBJECT

Strength of Materials

TODAY'S TOPIC

Bending & Shear Stresses

COUNTDOWN

102 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on the flexure formula, section modulus, T- and I-sections, beams of uniform strength, flitched beams and shear stress distributions. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

Which of the following is **not** an assumption of the simple bending (flexure) theory $\sigma/y = M/I = E/R$?

- (A) Plane sections remain plane after bending
- (B) The material is homogeneous and obeys Hooke's law
- (C) E is the same in tension and in compression
- (D) Bending stress varies parabolically across the depth

Q2 (NAT)

A rectangular beam 80 mm wide and 160 mm deep carries a bending moment of 8 kN·m about its horizontal axis. The maximum bending stress (in MPa, round off to two decimal places) is _____

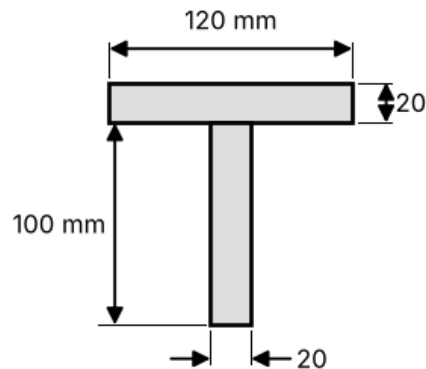
Q3 (MCQ)

The section modulus of a hollow circular section of outer diameter D and inner diameter d is

- (A) $\pi(D^4 - d^4)/(32D)$
- (B) $\pi(D^3 - d^3)/32$
- (C) $\pi(D^4 - d^4)/(64D)$
- (D) $\pi(D^4 - d^4)/(16D)$

Q4 (NAT)

The T-section shown (flange 120 mm × 20 mm, web 20 mm × 100 mm) is used with the flange on top and carries a sagging bending moment of 6 kN·m. The maximum **tensile** bending stress (in MPa, round off to one decimal place) is _____

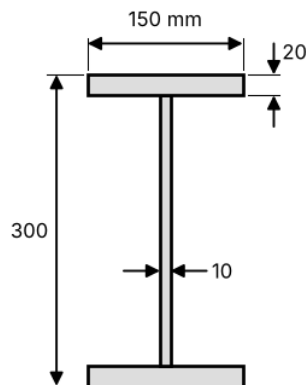
**Q5** (MCQ)

A solid circular beam of diameter 100 mm carries a shear force of 47.12 kN. The maximum shear stress in the section is nearly

- (A) 8 MPa (B) 6 MPa
(C) 9 MPa (D) 12 MPa

Q6 (NAT)

The symmetric I-section shown (flanges 150 mm × 20 mm, web 10 mm thick, overall depth 300 mm) carries a shear force of 100 kN. The maximum shear stress in the section (in MPa, round off to one decimal place) is _____

**Q7** (MCQ)

If the width of a rectangular beam is doubled and its depth is halved (same material), its moment-carrying capacity becomes

- (A) half (B) the same
(C) double (D) one-quarter

Q8 (NAT)

A steel wire of diameter 3 mm is to be bent into a circular arc. Taking $E = 210$ GPa, the smallest radius of curvature of the wire axis (in m) for which the bending stress does not exceed 210 MPa is _____

Q9 MCQ

The strongest rectangular beam (in bending) that can be cut from a circular log of diameter D has a depth-to-width ratio of

- (A) $\sqrt{2}$ (B) $\sqrt{3}$
(C) 2 (D) 1.5

Q10 NAT

A simply supported beam of span 4 m with a rectangular section 100 mm wide $\times$ 200 mm deep carries a single point load P at midspan. If the permissible bending stress is 12 MPa, the largest P (in kN) is _____

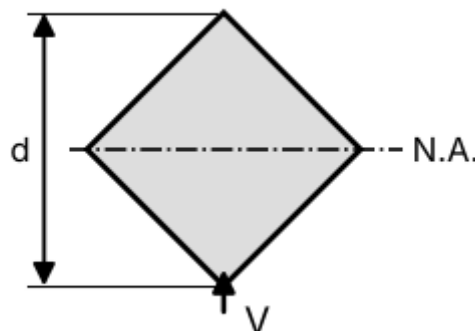
Q11 MCQ

In the flange of an I-beam carrying a vertical shear force, the horizontal shear stress (shear flow) along the flange width varies

- (A) linearly from zero at the flange tip to a maximum at the web (B) parabolically, maximum at the flange tip
(C) uniformly across the flange (D) as a cubic, maximum at the flange tip

Q12 NAT

A beam has a square cross-section placed with one diagonal vertical (as shown) and carries a vertical shear force. The ratio of the maximum shear stress to the average shear stress is _____

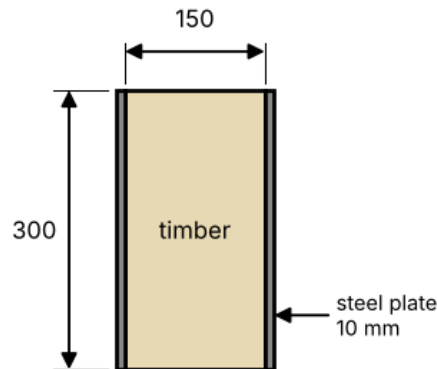
**Q13** MCQ

A simply supported beam carries a central point load. To be a beam of uniform strength with a **constant depth**, its width must vary along the span (from a support towards the centre)

- (A) linearly with distance from the support (B) with the square root of the distance
(C) with the square of the distance (D) it must stay constant

Q14 (NAT)

A flitched beam is made of a timber section 150 mm wide $\times$ 300 mm deep with a 10 mm $\times$ 300 mm steel plate bolted on each side (as shown). $E_{\text{steel}}/E_{\text{timber}} = 20$ and the permissible stresses are 8 MPa in timber and 150 MPa in steel. The moment of resistance of the beam (in kN·m, round off to two decimal places) is _____

**Q15** (MCQ)

At the extreme top and bottom fibres of a beam section carrying both bending moment and shear force,

- (A) bending stress is maximum and shear stress is zero (B) both stresses are maximum
(C) bending stress is zero and shear stress is maximum (D) both stresses are zero

Answer Key

Q	Type	Answer
Q1	MCQ	(D) Bending stress varies parabolically across the depth
Q2	NAT	23.44
Q3	MCQ	(A) $\pi(D^4 - d^4)/(32D)$
Q4	NAT	87.5
Q5	MCQ	(A) 8 MPa
Q6	NAT	38.1
Q7	MCQ	(A) half
Q8	NAT	1.5
Q9	MCQ	(A) $\sqrt{2}$
Q10	NAT	8
Q11	MCQ	(A) linearly from zero at the flange tip to a maximum at the web
Q12	NAT	1.125
Q13	MCQ	(A) linearly with distance from the support
Q14	NAT	61.88
Q15	MCQ	(A) bending stress is maximum and shear stress is zero

Detailed Solutions

Q1 Answer: (D) Bending stress varies parabolically across the depth

Concept: Plane sections remaining plane gives strain $\epsilon = y/R$, which is *linear* in y .

With Hooke's law and the same E in tension and compression, $\sigma = Ey/R$ is also linear in y , zero at the neutral axis.

A parabolic variation is the shape of the **shear** stress in a rectangular beam, not the bending stress.

Exam tip: Bending stress: linear, max at extreme fibres. Shear stress: parabolic (rectangle), max at the N.A.

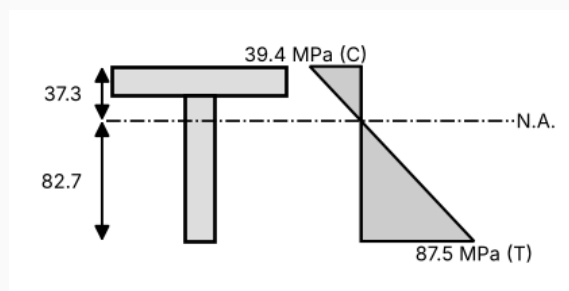
Q2 Answer: 23.44

Concept: $\sigma_{\max} = M/Z$ with $Z = bd^2/6$.

$$Z = 80 \times 160^2/6 = 3.413 \times 10^5 \text{ mm}^3$$

$$\sigma_{\max} = 8 \times 10^6 / 3.413 \times 10^5 = \mathbf{23.44 \text{ MPa}}$$
 (compression at top, tension at bottom for sagging).

Exam tip: Keep N and mm together: $\text{kN}\cdot\text{m} \times 10^6 = \text{N}\cdot\text{mm}$, and the answer is directly in MPa.

Q3 Answer: (A) $\pi(D^4 - d^4)/(32D)$ **Concept:** $Z = I/y_{\max}$, with $y_{\max} = D/2$. $I = \pi(D^4 - d^4)/64$, so $Z = [\pi(D^4 - d^4)/64]/(D/2) = \pi(D^4 - d^4)/(32D)$. $\pi(D^3 - d^3)/32$ is the common wrong answer: section moduli cannot be subtracted like moments of inertia.**Exam tip:** Subtract I's (same axis), never Z's.**Q4 Answer: 87.5****Concept:** Locate the centroid, find I about it (parallel-axis theorem), then $\sigma = My/I$ at the extreme tension fibre.Areas: flange 2400 mm^2 (centroid 10 mm from top), web 2000 mm^2 (centroid 70 mm from top). $\bar{y}$ from top = $(2400 \times 10 + 2000 \times 70)/4400 = 37.27 \text{ mm}$; bottom fibre is $120 - 37.27 = 82.73 \text{ mm}$ below the N.A. $I = 120 \cdot 20^3/12 + 2400 \cdot 27.27^2 + 20 \cdot 100^3/12 + 2000 \cdot 32.73^2 = 5.674 \times 10^6 \text{ mm}^4$ Sagging $\rightarrow$ tension at the bottom: $\sigma_t = 6 \times 10^6 \times 82.73/5.674 \times 10^6 = \mathbf{87.5 \text{ MPa}}$; compression at the top is only 39.4 MPa .**Exam tip:** For unsymmetric sections the two extreme stresses differ; always check which fibre governs.**Q5 Answer: (A) 8 MPa****Concept:** For a solid circle $\tau_{\max} = (4/3) \tau_{\text{avg}}$, at the neutral axis. $A = \pi \times 100^2/4 = 7854 \text{ mm}^2$; $\tau_{\text{avg}} = 47120/7854 = 6.0 \text{ MPa}$ $\tau_{\max} = (4/3) \times 6 = \mathbf{8 \text{ MPa}}$ **Exam tip:** Factors to remember: rectangle $3/2$, circle $4/3$, triangle $3/2$ (at mid-height), diamond $9/8$.

Q6 Answer: 38.1

Concept: $\tau = VQ/(I t)$; the maximum is at the neutral axis, where the width t is the web thickness.

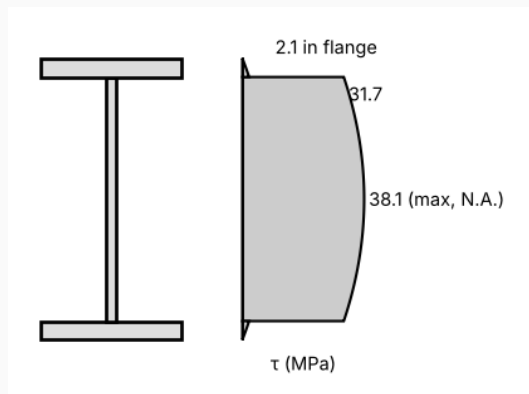
$$I = (150 \times 300^3 - 140 \times 260^3)/12 = 1.3245 \times 10^8 \text{ mm}^4$$

$$Q \text{ at N.A.} = \text{flange } 150 \times 20 \times 140 + \text{half-web } 10 \times 130 \times 65 = 420000 + 84500 = 504500 \text{ mm}^3$$

$$\tau_{\max} = 100000 \times 504500 / (1.3245 \times 10^8 \times 10) = \mathbf{38.1 \text{ MPa}}$$

At the flange-web junction τ jumps from 2.1 MPa (in the flange, $t = 150$) to 31.7 MPa (in the web, $t = 10$). The web carries about 93 % of the shear.

Exam tip: A quick check: $V/(\text{web area}) = 100000/(300 \times 10) = 33.3 \text{ MPa}$ is close to $\tau_{\max}$ for I-beams.

**Q7 Answer: (A) half**

Concept: $M = \sigma Z$ and $Z = bd^2/6 \propto bd^2$.

New $Z \propto (2b)(d/2)^2 = bd^2/2 \rightarrow$ capacity becomes **half**.

Depth matters more than width because it enters squared.

Exam tip: Stiffness ($I \propto bd^3$) would fall to one-quarter in the same change.

Q8 Answer: 1.5

Concept: From $\sigma/y = E/R \rightarrow R = E y_{\max}/\sigma$.

$$y_{\max} = 1.5 \text{ mm}$$

$$R = 210000 \times 1.5/210 = 1500 \text{ mm} = \mathbf{1.5 \text{ m}}$$

Exam tip: If a drum radius is asked instead, subtract the wire radius: $1500 - 1.5 = 1498.5 \text{ mm}$.

Q9 Answer: (A) $\sqrt{2}$

Concept: Maximise $Z = bh^2/6$ with $b^2 + h^2 = D^2$.

$$Z \propto b(D^2 - b^2); dZ/db = D^2 - 3b^2 = 0 \rightarrow b = D/\sqrt{3}, h = D\sqrt{2/3}$$

$$h/b = \sqrt{2}$$

Exam tip: The *stiffest* beam ($\max I \propto bh^3$) gives $h/b = \sqrt{3}$. Do not mix them up.

Q10 Answer: 8

Concept: $M_{\max} = PL/4$ at midspan; set $M_{\max} = \sigma Z$.

$$Z = 100 \times 200^2/6 = 6.667 \times 10^5 \text{ mm}^3; \sigma Z = 12 \times 6.667 \times 10^5 = 8 \times 10^6 \text{ N}\cdot\text{mm} = 8 \text{ kN}\cdot\text{m}$$

$$PL/4 = P \times 4/4 = P \rightarrow \mathbf{P = 8 \text{ kN}}$$

Exam tip: Self-weight is ignored unless the question gives the density.

Q11 Answer: (A) linearly from zero at the flange tip to a maximum at the web

Concept: In the flange, Q of the part from the tip to a cut at distance s is $(s t_f) \times (h/2)$, which grows linearly with s .

$\tau = VQ/(I t_f) \propto s \rightarrow$ zero at the free tip, maximum where the flange meets the web.

This flange shear flow is what locates the shear centre of channel sections.

Exam tip: In the web the variation is parabolic; in thin flanges it is linear.

Q12 Answer: 1.125

Concept: $\tau = VQ/(I b)$; for the diamond the width b falls towards the N.A. faster than Q grows near it, so τ peaks away from the N.A.

With half-diagonal h : $b = 2(h - y)$, $Q = (h - y)^2(h + 2y)/3$, so $\tau \propto (h - y)(h + 2y)$; $d\tau/dy = 0$ gives $y = h/4 = d/8$ from the N.A. (d = full diagonal)

$\tau_{\max} = (9/8) \tau_{\text{avg}} = \mathbf{1.125 \tau_{\text{avg}}}$; at the N.A. itself $\tau = \tau_{\text{avg}}$.

Exam tip: Diamond and triangle are the two GATE favourites where $\tau_{\max}$ is not at the N.A.

Q13 Answer: (A) linearly with distance from the support

Concept: Uniform strength: $\sigma = 6M(x)/(b h^2)$ is the same at every section.

$M(x) = (P/2)x$ for $0 \leq x \leq L/2$; with h fixed, $b \propto M \propto x$ (linear).

With constant width instead, $h^2 \propto x \rightarrow h \propto \sqrt{x}$.

Exam tip: Leaf springs are a practical form of such a constant-depth, varying-width beam.

Q14 Answer: 61.88

Concept: Transform the steel into equivalent timber (width $\times m$); strains are equal at the same depth, so $\sigma_s = m \sigma_t$ at the same fibre.

Equivalent timber width = $150 + 20 \times (2 \times 10) = 550$ mm; $Z_{\text{eq}} = 550 \times 300^2/6 = 8.25 \times 10^6$ mm³

If timber reaches 8 MPa, steel would be $20 \times 8 = 160$ MPa > 150 MPa. So steel governs: $\sigma_t = 150/20 = 7.5$ MPa at the extreme fibre.

$M = 7.5 \times 8.25 \times 10^6 = 61.875 \times 10^6$ N·mm = **61.88 kN·m**

Exam tip: Always check both materials; the one that reaches its limit first decides M .

Q15 Answer: (A) bending stress is maximum and shear stress is zero

Concept: $\sigma = My/I$ is largest at $y_{\max}$.

The top and bottom surfaces are free of load, so by complementary shear the transverse shear stress there is **zero**.

At the N.A. the situation is reversed: $\sigma = 0$ and τ is (usually) maximum.

Exam tip: This is why principal-stress questions on beams take the extreme fibre as uniaxial.

What next?

Day 8 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



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Next, Day 9/90: General Aptitude, Verbal Ability. Comment **GATE90** on our reel to get all links.

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